

Daily Progress Report Day 1

Project Name: MediFlow CRM — Doctor / Clinic Practice Management System

Date: September 14, 2026

Developer/Reporter: Debarati Gantait, Anwesha Manna, Indranil Dutta

Reporting Period: Daily

1. Summary of Today's Activities

Today, the team conducted an in-depth review and technical analysis of the 18-page system specification document for the **MediFlow Clinic CRM System**. As our team is building a full-featured, secure medical practice management system using **PHP and the Laravel framework**, we dedicated the day to extracting functional requirements, defining domain models, dividing core architectural responsibilities among all three members, configuring the local development environment (PHP 8.3 CLI, Composer 2.10, MySQL/SQLite database engine), and mapping out the unified clinical and financial workflow.

2. Key Discussions & Decisions

A. Architecture & Workflow Design

- **System Architecture:** Agreed on a clean, scalable Laravel MVC architecture utilizing Eloquent ORM, Form Requests, Policies, Middleware, and Blade templates.
- **Central Practice Lifecycle Workflow:** Clarified and finalized the end-to-end patient journey to eliminate redundant data entry across clinical and front-desk staff: Login → Dashboard → Patient → Appointment → Check-In → Consultation → Prescription → Invoice → Payment → Completed
- **Database Schema Planning:** Outlined the normalized relational schema for 12+ core domain entities (users, roles_permissions, patients, patient_notes, patient_documents, appointments, consultations, prescriptions, prescription_items, invoices, invoice_items, payments, settings) with strict foreign key integrity and cascading deletes.
- **Dual Operational Workflows:**
 1. **Receptionist & Scheduling Flow:** Patient search/registration → Appointment booking across List & Calendar (Day/Week/Month) → Physical Check-In upon patient arrival.
 2. **Doctor & Clinical Flow:** Start Consultation → Review past medical profile → Log Symptoms, Diagnosis, Notes, Treatment → Author structured multi-line digital Prescription → Trigger itemized Billing Invoice → Record multi-mode Payment & complete visit.

B. Tech Stack Learning & Setup

- **PHP 8.3 & Laravel Foundations:** Verified local PHP 8.3 and Composer toolchain, configured system environment PATH, set up database connectivity, and established directory structures.

- **Libraries & Packages:** Selected **DomPDF** (barryvdh/laravel-dompdf) for pixel-accurate digital prescription and billing invoice PDF generation; evaluated **Chart.js** for Dashboard revenue and practice metric visualizations.
- **Security & RBAC Architecture:** Defined 6 user roles (**Admin / Doctor, Clinic Manager, Receptionist, Nurse / Assistant, Accountant, Other Staff**) governed by a 22-permission configurable access matrix. Enforced that sensitive permissions (record deletions, practice settings, staff directory management) remain restricted by default.

3. Members Role

Team Member	Assigned Role	Module & Responsibility Breakdown
<i>Anwisha Manna(Backend & Database)</i>	Backend Architecture & Database Lead	• Master Database Schema design, Migrations & Eloquent Models • User Authentication, Session handling & Subaccounts RBAC engine (6 Roles & 22 Configurable Permissions) • Patient Directory & 9-Section Profile Hub (Demographics, Medical History, Document Attachments, Notes) • Practice Settings Repository (Doctor Profile, Clinic branding, Appointment rules, Security)
Indranil Dutta (<i>Full-Stack / Clinical Workflow Lead</i>)	Clinical Engine & Workflow Specialist	• Appointment Scheduling System (List view, Day/Week/Month calendar views, 7 lifecycle statuses) • Clinical Consultation Encounter Workspace (Symptoms, Diagnosis, Medical Notes, Treatment) • Digital Prescription Builder (Structured medicine dosage, frequency, duration, route, timing, instructions) • PDF Generation & Printing Engine for Prescriptions via DomPDF
Debarati Gantait (<i>Frontend, Billing & Reporting Lead</i>)	Billing, Analytics & UI/UX Lead	• Invoice & Billing Engine (Itemized services, tax/discount calculations, 6 invoice statuses) • Multi-Mode Payment Recording (Cash, Card, Bank Transfer, UPI, Other) • Central Practice Cockpit Dashboard (9 KPI cards, Today's queue, Monthly Revenue trend chart) • Comprehensive Reporting Engine (Patient, Appointment, Financial, and Prescription reports)

		with export/print) • Cross-Device Responsive Layout (Sidebar mobile drawer, responsive table-to-card transformations)
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4. Roadblocks & Knowledge Gathering Strategy

- **Area of Focus:** Establishing a seamless, single-action data bridge between Clinical Consultations, Prescriptions, and Invoicing without requiring the practitioner to re-type diagnosis or patient details, alongside adapting multi-column clinical tables into mobile-friendly cards.
 - **Action Taken:** Designed an integrated consultation state machine linking consultation_id directly to generated prescriptions and invoices; created responsive UI component specifications following Section 15 of the requirements document for off-canvas mobile drawer navigation and dynamic touch-friendly card layouts.
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Submitted By GROUP A

(Anwesha Manna, Debarati Gantait, Indranil Dutta)